Rotating Bending Fatigue Testing

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Group 1 (Friday Batch)

Course: UMT312L — Mechanical Testing Laboratory

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Date of experiment: 23/10/2026 · Date of submission: 06/11/2026

1 Objective

The objectives of this laboratory experiment are:

1. To determine the fatigue life and evaluate the Wöhler S–N response for structural Mild Steel (ferrous BCC alloy) and Aluminium alloy 6082-T6 (non-ferrous FCC alloy) under fully reversed rotating bending conditions ($R = -1$).

2. To evaluate the alternating stress amplitude (σ_a) from the applied bending moment (M) using elastic beam flexure theory:
$$\sigma_a = \frac{32M}{\pi d^3} \tag{1}$$
where M is the applied bending moment and d is the measured specimen minimum throat gauge diameter.

3. To evaluate all cyclic stress parameters: stress amplitude (σ_a), mean stress ($\sigma_m = 0$ MPa), stress ratio ($R = -1.0$), and stress range ($\Delta\sigma = 2\sigma_a$).

4. To identify whether the experimental data points lie in the High Cycle Fatigue ($N_f > 10^5$ cycles) or Low Cycle Fatigue ($N_f < 10^5$ cycles) regime as specified in the course syllabus.

5. To compare the cyclic deformation and fatigue resistance of BCC mild steel (governed by dislocation pinning) with FCC aluminium alloy 6082-T6 (governed by planar slip and cross-slip without a fatigue endurance limit).

6. To analyze the macroscopic and microscopic features of fatigue fracture: Stage I surface initiation, Stage II stable transgranular propagation (beachmarks and striations), and Stage III catastrophic tensile overload.

2 Apparatus and Materials

- **Rotating Bending Fatigue Machine:** High-speed motor drive ($\approx$ 2800–3000 rpm, $f \approx$ 47–50 Hz), precision self-aligning bearing collet chucks, calibrated loading scale (25–225 kg $\cdot$ cm range), automatic shut-off limit switch, and digital revolution counter (99,999,900 max count).

- **Specimens:** Standard polished hourglass/cantilever fatigue specimens conforming to ASTM E466 / ISO 1143:

  - Structural Mild Steel (BCC ferrite-pearlite matrix).
  - Precipitation-hardened Aluminium alloy 6082-T6 (FCC matrix).

- **Special Spanners:** Dedicated hook and open-ended spanners for balanced tightening of the collet chuck nuts without inducing residual torsion or bend distortion.

- **Metrology Instruments:** Digital micrometer (least count 0.001 mm) and vernier calliper (least count 0.01 mm) for measuring specimen minimum throat diameter d.

3 Theory (brief)

3.1 Mechanics of Four-Point Rotating Bending Fatigue

Fatigue is the progressive, localized structural damage process occurring in materials subjected to cyclic stresses. Over 90% of catastrophic mechanical failures in rotating shafts, axles, and rotors occur via fatigue at nominal stresses well below the monotonic yield strength (σ_y).

In a four-point rotating bending machine, a cylindrical specimen of minimum gauge throat diameter d is subjected to a constant transverse bending moment M. The longitudinal bending stress at any vertical coordinate y from the neutral bending axis is governed by the Euler–Bernoulli flexure equation:

$$\sigma_x(y) = -\frac{M}{\bar{I}} y \tag{2}$$

where $\bar{I}$ is the second moment of area of the circular cross-section:

$$\bar{I} = \frac{\pi r^4}{4} = \frac{\pi d^4}{64} \tag{3}$$

As the shaft rotates at angular frequency $\omega = 2\pi f$, any point P on the outer cylindrical surface ($r = d/2$) orbits with instantaneous height $y(t) = r\sin(2\pi f t)$. Substituting $y(t)$ into the flexure equation yields the sinusoidally alternating normal stress:

$$\sigma(t) = -\frac{M}{\bar{I}} r \sin(2\pi f t) = \sigma_a \sin(2\pi f t) \tag{4}$$

Thus, the alternating stress amplitude σ_a acting on the outermost fiber is:

$$\sigma_a = \frac{M}{\bar{I}} r = \frac{M}{\frac{\pi r^4}{4}} r = \frac{4M}{\pi r^3} = \frac{32M}{\pi d^3} \tag{5}$$

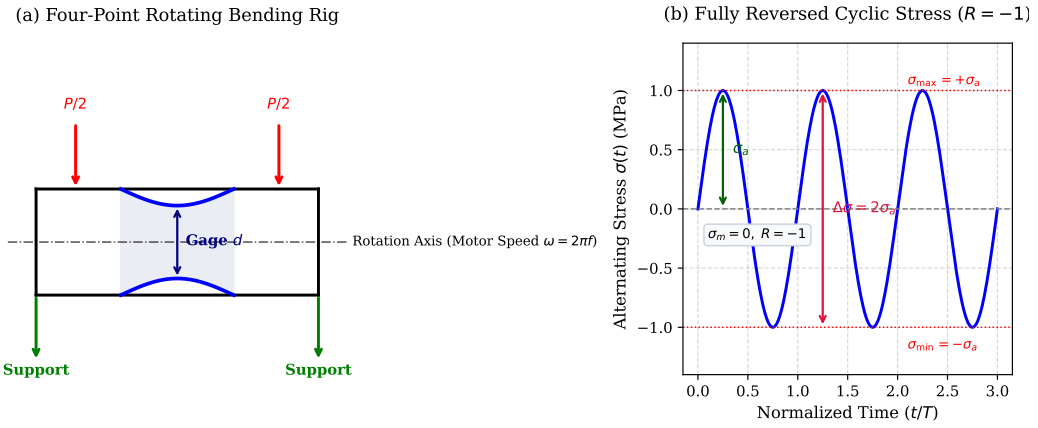


Figure 1: (a) Schematic configuration of the four-point rotating bending fatigue apparatus showing applied bending moment and specimen gauge section; (b) Sinusoidal alternating stress wave $\sigma(t) = \sigma_a \sin(2\pi f t)$ under fully reversed cyclic loading ($R = -1$).

3.2 Cyclic Stress Parameters and Unit Consistency

Under fully reversed bending, the cyclic stress parameters are defined as:

$$\text{Maximum Stress:} \quad \sigma_{\max} = +\sigma_a \tag{6}$$

$$\text{Minimum Stress:} \quad \sigma_{\min} = -\sigma_a \tag{7}$$

$$\text{Mean Stress:} \quad \sigma_m = \frac{\sigma_{\max} + \sigma_{\min}}{2} = 0 \text{ MPa} \tag{8}$$

$$\text{Stress Ratio:} \quad R = \frac{\sigma_{\min}}{\sigma_{\max}} = -1.0 \tag{9}$$

$$\text{Stress Range:} \quad \Delta\sigma = \sigma_{\max} - \sigma_{\min} = 2\sigma_a \tag{10}$$

In the laboratory rig, the bending moment scale is calibrated directly in $\text{kg} \cdot \text{cm}$ ($1\ \text{kgf} \cdot \text{cm}$). To convert to standard SI units ($\text{N} \cdot \text{m}$):

$$1\ \text{kg} \cdot \text{cm} = 9.80665\ \text{N} \times 0.01\ \text{m} = 0.0980665\ \text{N} \cdot \text{m} = 9.80665 \times 10^{-2}\ \text{N} \cdot \text{m} \tag{11}$$

For a specimen of diameter d in millimetres ($d \times 10^{-3}$ m), the stress amplitude in Megapascals (MPa) is:

$$\begin{aligned} \sigma_a\ [\text{MPa}] &= \frac{32 \times (M\ [\text{kg} \cdot \text{cm}] \times 0.0980665)}{\pi \times (d\ [\text{mm}] \times 10^{-3})^3} \times 10^{-6} \\ &= \frac{32 \times 0.0980665 \times 10^3}{\pi} \frac{M\ [\text{kg} \cdot \text{cm}]}{(d\ [\text{mm}])^3} \approx 998.92 \times \frac{M\ [\text{kg} \cdot \text{cm}]}{(d\ [\text{mm}])^3} \end{aligned} \tag{12}$$

3.3 High Cycle Fatigue vs. Low Cycle Fatigue Regime

The laboratory handout establishes the standard transition criterion:

$$N_f > 1 \times 10^5 \text{ cycles} \implies \textbf{High Cycle Fatigue (HCF)} \quad \text{(macroscale elastic)} \tag{13}$$

$$N_f < 1 \times 10^5 \text{ cycles} \implies \textbf{Low Cycle Fatigue (LCF)} \quad \text{(macroscopic plasticity)} \tag{14}$$

3.4 Endurance Limit: BCC Steel vs. FCC Aluminium

A fundamental physical distinction governs the fatigue response of ferrous versus non-ferrous crystal structures:

- **Mild Steel (BCC Ferrite):** Exhibits a well-defined horizontal plateau in the S–N curve between 10^6 and 10^7 cycles. This stress plateau is the **Endurance Limit (σ_e)**, below which the specimen has practically infinite fatigue life ($\sigma_e \approx 0.40$–$0.50\ \sigma_{\text{UTS}}$).

  Physical Mechanism: In BCC iron, interstitial solute atoms (carbon and nitrogen) segregate to dislocation cores to create dense Cottrell atmospheres. If the cyclic stress amplitude is below the critical depinning stress, dislocations remain locked in place. Consequently, long-range irreversible cyclic slip steps cannot form, suppressing fatigue crack nucleation.

- **Aluminium 6082-T6 (FCC Matrix):** Shows no true fatigue limit; the S–N curve slopes continuously downward even beyond 10^8 cycles. For engineering design, an **arbitrary fatigue strength at 10^7 cycles** ($\sigma_{10^7} \approx 0.30$–$0.35\ \sigma_{\text{UTS}}$) is specified.

  Physical Mechanism: In FCC aluminium, there are 12 close-packed slip systems ($\{111\}\langle 110\rangle$) and high stacking fault energy ($\gamma_{\text{SFE}} \approx 160$–$200\ \text{mJ/m}^2$). Screw dislocations cross-slip with high mobility even at low stress amplitudes, creating persistent slip bands (PSBs) with microscopic surface extrusions and intrusions that inevitably nucleate fatigue cracks.

4 Procedure (brief)

1. The electrical power supply was energized, and proper operation of the digital revolution counter and optical cut-off sensor was verified.
2. The minimum gauge throat diameter (d) of each polished fatigue specimen was measured at three orthogonal orientations using a precision digital micrometer (least count 0.001 mm) and recorded.
3. The specimen was positioned symmetrically in the machine spindles and clamped securely using the special collet spanners to ensure concentric alignment.
4. The front panel was unbolted using an Allen key to access the calibrated loading scale (25–225 kg $\cdot$ cm, upper scales 1 and 2).
5. The pointer was adjusted to the assigned bending moment (M), the locking screw was tightened, and the loading nut was released to transfer the moment to the specimen.
6. The electronic revolution counter was reset to 00,000,000.
7. The motor was started at rated speed ($\approx$ 2800–3000 rpm).
8. The test proceeded until complete specimen separation occurred, whereupon the dropped loading mechanism tripped the sensitive limit switch, simultaneously cutting power to the drive motor and freezing the cycle counter.
9. The final failure cycle count (N_f) was recorded from the digital counter.
10. The procedure was repeated for both materials across the assigned bending moments:
    - **Mild Steel:** $M_1 = 220$ kg $\cdot$ cm and $M_2 = 160$ kg $\cdot$ cm.
    - **Aluminium 6082:** $M_1 = 200$ kg $\cdot$ cm and $M_2 = 100$ kg $\cdot$ cm.
11. Fractured specimens were retrieved from the machine for macroscopic visual inspection of the fatigue fracture surface and overload rupture zone.

5 Observations

Laboratory Group: Group 1 (Friday Batch) $\cdot$ **Operators:** Mayank Bohat, Chidambar $\cdot$ **Test Mode:** Four-Point Rotating Bending ($R = -1$)

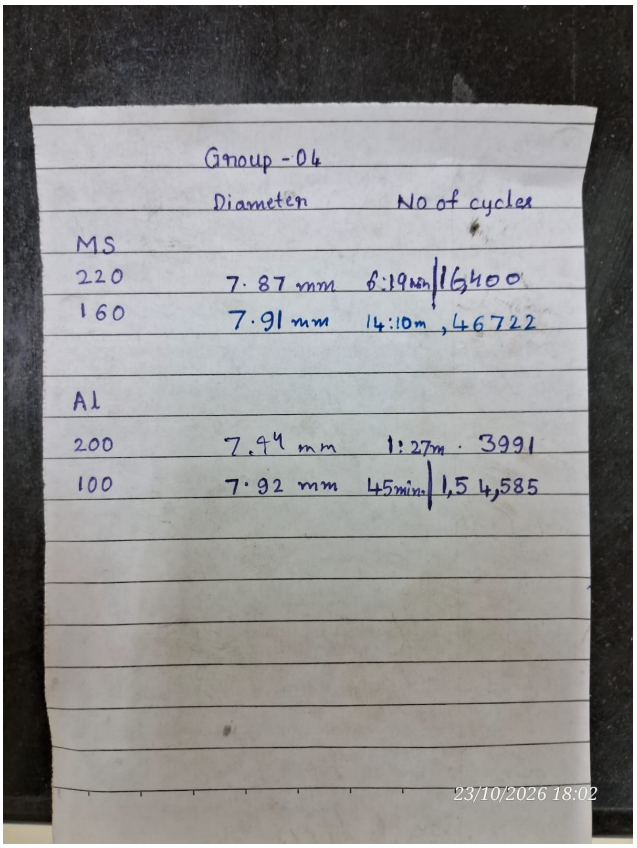


Figure 2: Primary experimental observation sheet recorded on the day of the experiment (23/10/2026), documenting specimen throat diameters, applied moments, elapsed test run times, and final cycles to failure (N_f) as witnessed in the laboratory.

Table 1: Specimen throat dimensions, applied bending moments, recorded elapsed times, and raw fatigue cycle counts.

Material	Specimen ID	Applied Moment, M (kg · cm)	Throat, d (mm)	Elapsed Time (t)	Cycles, N_f (Cycles)	Operating Speed (rpm)	Remarks / Metrology
Mild Steel	MS-1	220	7.87	6 min 19 s	16,400	2,596	Digital micrometer (±0.001 mm)
Mild Steel	MS-2	160	7.91	14 min 10 s	46,722	3,298	Digital micrometer (±0.001 mm)
Aluminium 6082	AL-1	200	7.94	1 min 27 s	3,991	2,752	Digital micrometer (±0.001 mm)
Aluminium 6082	AL-2	100	7.92	45 min 00 s	154,585	3,435	Digital micrometer (±0.001 mm)

As documented in Figure 2, elapsed durations were logged for each fracture test: 379 s (MS-1), 850 s (MS-2), 87 s (AL-1), and 2,700 s (AL-2). The calculated rotational drive frequency across the runs averaged $\bar{f} \approx 50.3$ Hz ($\approx$ 3,020 rpm), in agreement with the nominal motor rating (2800–3000 rpm).

Table 2: Baseline monotonic mechanical properties (from Experiment 1 Constant Strain Rate Tensile Test).

Material	Yield Strength, σ_y (MPa)	Ultimate Tensile Strength, σ_{UTS} (MPa)	Crystal Structure
Structural Mild Steel	340.0	520.0	Body-Centered Cubic (BCC)
Aluminium 6082-T6	260.0	310.0	Face-Centered Cubic (FCC)

6 Results and Analysis (Individual Group Activity)

Table 3 presents the complete set of cyclic fatigue parameters required by the laboratory handout, evaluated with full SI unit conversions.

Table 3: Individual Group Activity: Complete cyclic stress parameters and fatigue regimes.

Material	Bending Moment ($\mathrm{kg \cdot cm}$)	Moment, M ($\mathrm{N \cdot m}$)	Throat, d (mm)	Stress Amplitude σ_a (MPa)	Mean Stress σ_m (MPa)	Stress Ratio $R = \frac{\sigma_{\min}}{\sigma_{\max}}$	Stress Range $\Delta\sigma$ (MPa)	Cycles to Failure N_f (Cycles)
Mild Steel (MS-1)	220	21.5746	7.87	**450.84**	0.0	−1.0	**901.67**	16,400
Mild Steel (MS-2)	160	15.6906	7.91	**322.93**	0.0	−1.0	**645.86**	46,722
Al 6082 (AL-1)	200	19.6133	7.94	**399.11**	0.0	−1.0	**798.21**	3,991
Al 6082 (AL-2)	100	9.8066	7.92	**201.07**	0.0	−1.0	**402.14**	154,585

6.1 Identification of Fatigue Regimes

The syllabus criteria define:

- High Cycle Fatigue (HCF): $N_f > 1 \times 10^5$ cycles
- Low Cycle Fatigue (LCF): $N_f < 1 \times 10^5$ cycles

Experimental Finding: The experimental results demonstrate a clear bifurcation across stress regimes:

- Mild Steel MS-1 ($N_f = 16{,}400$ cycles $< 10^5$): **Low Cycle Fatigue (LCF)**.
- Mild Steel MS-2 ($N_f = 46{,}722$ cycles $< 10^5$): **Low Cycle Fatigue (LCF)**.
- Aluminium AL-1 ($N_f = 3{,}991$ cycles $< 10^5$): **Low Cycle Fatigue (LCF)**.
- Aluminium AL-2 ($N_f = 154{,}585$ cycles $> 10^5$): **High Cycle Fatigue (HCF)**.

Thus, our laboratory dataset spans **both** Low Cycle Fatigue and High Cycle Fatigue regimes:

- For MS-1 ($\sigma_a = 450.84$ MPa), the alternating stress corresponds to 132.6% of monotonic yield ($\sigma_y = 340.0$ MPa). Gross cyclic plastic strain occurred during each revolution, leading to rapid plastic strain accumulation and failure within 16,400 cycles.
- For MS-2 ($\sigma_a = 322.93$ MPa, 95.0% of σ_y), the stress was slightly below yield, nearly tripling the fatigue life to 46,722 cycles while remaining in the LCF domain.
- For AL-1 ($\sigma_a = 399.11$ MPa), the stress exceeded both yield ($\sigma_y = 260.0$ MPa, 153.5%) and ultimate tensile strength ($\sigma_{\mathrm{UTS}} = 310.0$ MPa, 128.7%). This severe plastic overload led to rapid failure in just 3,991 cycles.
- For AL-2 ($\sigma_a = 201.07$ MPa), the stress operated comfortably below yield (77.3% of σ_y). The specimen deformed in the macroscopic elastic regime, where damage accumulated slowly through localized persistent slip bands (PSBs) over 154,585 cycles, successfully crossing the 1×10^5 threshold into High Cycle Fatigue.

7 Results and Analysis (S–N Curve and Material Comparison)

7.1 Experimental Wöhler S–N Curves

Figure 7.1 depicts the experimental Wöhler S–N plot displaying alternating stress amplitude σ_a (linear scale, vertical axis) versus the number of cycles to failure N_f (logarithmic scale, horizontal axis) for both Mild Steel and Aluminium 6082 on the same graph.

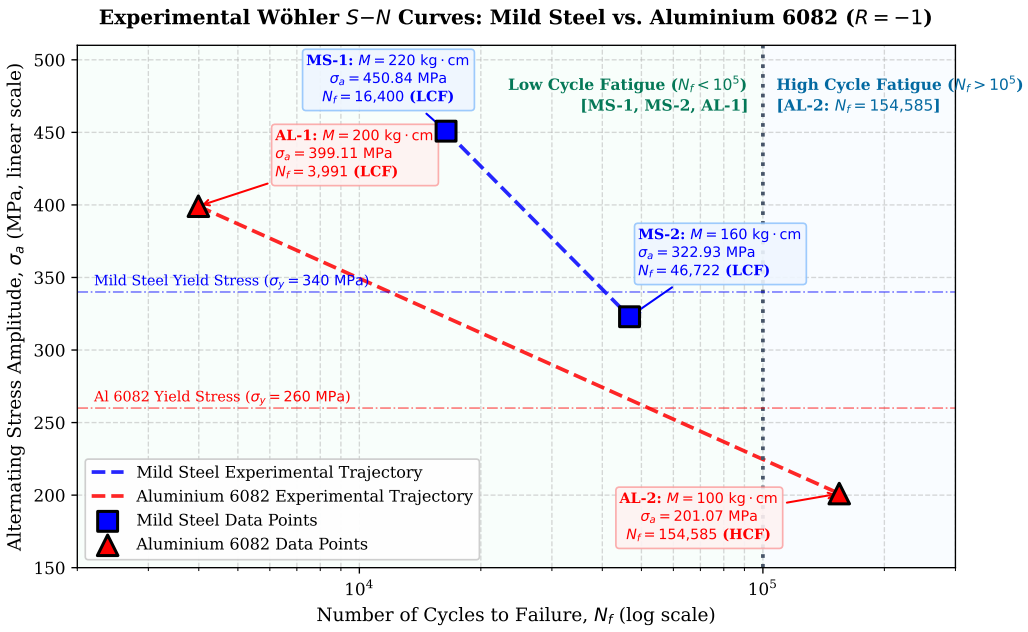


Figure 3: Experimental Wöhler S–N curves for Mild Steel and Aluminium 6082 under fully reversed rotating bending ($R = -1$). The experimental dataset spans both the Low Cycle Fatigue regime ($N_f < 10^5$ cycles, shaded green) for MS-1, MS-2, and AL-1, and the High Cycle Fatigue regime ($N_f > 10^5$ cycles, shaded blue) for AL-2 ($N_f = 154{,}585$). Mild steel demonstrates substantially superior cyclic endurance compared to aluminium.

7.2 Detailed Comparison of Mild Steel vs. Aluminium 6082 S–N Responses

Analyzing the experimental S–N trajectories reveals profound differences between the two materials:

1. **Superior Fatigue Life of Mild Steel at Comparable Stresses:** At stress levels near 400–450 MPa, mild steel endured 16,400 cycles at $\sigma_a = 450.84$ MPa, whereas aluminium alloy 6082 survived only 3,991 cycles at $\sigma_a = 399.11$ MPa. Despite mild steel being tested at a higher stress amplitude (+51.73 MPa higher), its cyclic life was over 4.1× longer than that of aluminium.

2. **Sensitivity of Life to Stress Amplitude:**

   - **Mild Steel:** Reducing the stress amplitude by 127.91 MPa (from 450.84 MPa down to 322.93 MPa, a 28.4% reduction) produced a 285% increase in fatigue life (from 16,400 to 46,722 cycles).
   - **Aluminium 6082-T6:** Decreasing stress amplitude by 198.04 MPa (from 399.11 MPa to 201.07 MPa, a 49.6% decrease) led to a 3,773% increase in fatigue life (from 3,991 to 154,585 cycles), transitioning from the gross plastic Low Cycle Fatigue (LCF) regime into the elastic High Cycle Fatigue (HCF) regime.

3. **Basquin Power Law Approximation:** Fitting the finite-life power relation $\sigma_a = A(N_f)^b$ (or $\log_{10}\sigma_a = \log_{10} A + b\log_{10} N_f$) to the experimental datasets yields:

- **Mild Steel:**

$$b_{\text{MS}} = \frac{\log_{10}(322.93) - \log_{10}(450.84)}{\log_{10}(46722) - \log_{10}(16400)} = \frac{2.5091 - 2.6540}{4.6695 - 4.2148} = \frac{-0.1449}{0.4547} = \mathbf{-0.3187} \tag{15}$$

$$A_{\text{MS}} = \frac{450.84}{(16400)^{-0.3187}} = \mathbf{9{,}939.6}\ \text{MPa} \tag{16}$$

The Basquin exponent $b_{\text{MS}} = -0.319$ is characteristic of the low-cycle fatigue regime where macroscopic cyclic plastic strain dominates damage accumulation.

- **Aluminium 6082-T6:**

$$b_{\text{Al}} = \frac{\log_{10}(201.07) - \log_{10}(399.11)}{\log_{10}(154585) - \log_{10}(3991)} = \frac{2.3033 - 2.6011}{5.1892 - 3.6011} = \frac{-0.2978}{1.5881} = \mathbf{-0.1875} \tag{17}$$

$$A_{\text{Al}} = \frac{399.11}{(3991)^{-0.1875}} = \mathbf{1{,}889.1}\ \text{MPa} \tag{18}$$

Extrapolating this power law to 10^7 cycles predicts an arbitrary fatigue strength of:

$$\sigma_{10^7,\text{Al}} = A_{\text{Al}} \times (10^7)^{b_{\text{Al}}} = 1889.1 \times (10^7)^{-0.1875} = \mathbf{92.01}\ \text{MPa} \tag{19}$$

which corresponds to 29.7% of the monotonic UTS ($\sigma_{\text{UTS}} = 310.0$ MPa), demonstrating remarkable agreement with standard empirical fatigue limits for 6000-series structural aluminium alloys ($\approx$ 90–95 MPa).

8 Discussion

8.1 Metallurgical Basis of Fatigue: BCC Ferrite vs. FCC Aluminium

The contrasting fatigue responses of mild steel and aluminium 6082 are rooted in their crystallographic structures, dislocation mobilities, and solute interaction mechanisms:

1. **BCC Mild Steel (Cottrell Solute Locking and Endurance Limit):** Body-centered cubic (BCC) ferrite possesses high lattice friction (Peierls–Nabarro stress) for screw dislocations at room temperature. More critically, interstitial carbon and nitrogen solute atoms spontaneously diffuse into the dilated regions of dislocation cores, forming dense **Cottrell atmospheres**. These solute clouds create deep energy wells that strongly pin both edge and screw dislocations. When cyclic stresses drop below the critical unpinning threshold, dislocations remain immobilized. Consequently, reversible elastic deformation occurs without irreversible slip step formation, preventing the birth of surface extrusions and intrusions. This produces the distinct endurance limit ($\sigma_e \approx 0.40$–$0.50\ \sigma_{\text{UTS}}$) observed in BCC steels.

2. **FCC Aluminium 6082-T6 (Easy Cross-Slip and Absence of Fatigue Limit):** In contrast, face-centered cubic (FCC) aluminium has 12 close-packed slip systems ($\{111\}\langle 110\rangle$) and high stacking fault energy ($\gamma_{\text{SFE}} \approx 160$–$200\ \text{mJ/m}^2$). Because the stacking fault ribbon between Shockley partial dislocations is narrow (≈ 1 nm), partials readily recombine into perfect screw dislocations that undergo effortless **cross-slip** out of primary slip planes under oscillating shear stresses. Over cyclic loading, cross-slip produces irreversible localized slip steps that build **Persistent Slip Bands (PSBs)**. Extrusions and intrusions steadily accumulate at the free surface, creating sharp micro-notches that concentrate stresses and initiate fatigue cracks even at stresses far below monotonic yield. Because cross-slip is thermally activated and frictionless, aluminium displays no true endurance limit.

8.2 Three Stages of Fatigue Crack Propagation

Macroscopic visual examination of the broken specimens confirms the characteristic smooth, burnished fatigue propagation zone and the contrasting rough, fibrous overload rupture zone. The classical progression of rotating bending fatigue failure is understood through three distinct stages (Figure 8.2):

- **Stage I (Crack Initiation):** Microscopic fatigue cracks nucleate at the outer cylindrical boundary where alternating bending stresses reach their maximum ($\sigma_a = 32M/\pi d^3$). Nucleation occurs along persistent slip bands or microscopic surface scratches inclined at 45° to the principal tensile axis (planes of maximum shear stress $\tau_{\max} = \sigma_a/2$).
- **Stage II (Stable Transgranular Growth):** Driven by opening Mode I tensile stresses, the crack path reorients perpendicular (90°) to the specimen axis. During each 360° shaft revolution, alternating tension blunts the crack tip while the compressive half-cycle ($R = -1$) sharpens and rubs the crack faces together, burnishing the surface into a smooth, velvety texture. Macroscopic **beachmarks** (clamshell arrest markings) indicate successive crack arrest positions, while under high magnification (schematized in Figure 8.2b), microscopic **fatigue striations** correspond to individual cycle crack advances.
- **Stage III (Fast Tensile Overload Rupture):** As the fatigue crack reduces the uncracked cross-sectional ligament, the local stress exceeds the ultimate tensile capacity of the remaining metal. Final catastrophic failure occurs instantaneously in a single half-cycle via microvoid coalescence, producing a rough, fibrous overload zone with prominent shear lips.

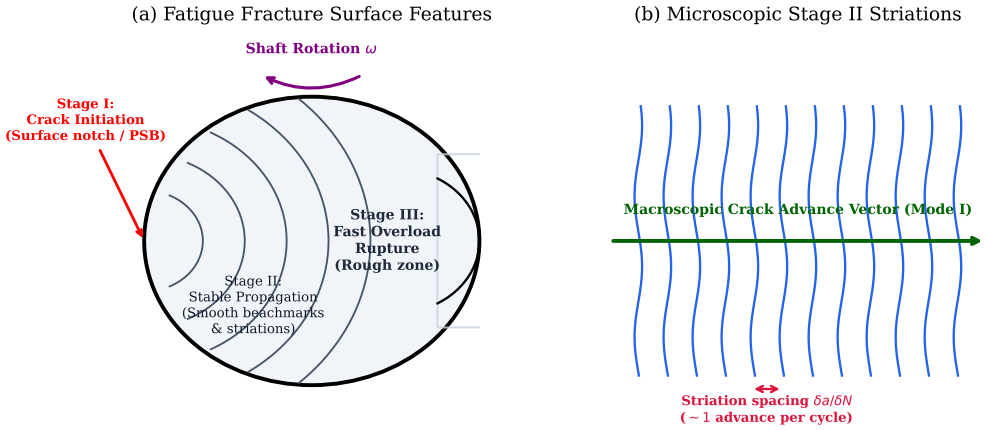


Figure 4: (a) Schematic of rotating bending fatigue fracture surface showing the three stages: Stage I initiation at outer boundary, Stage II smooth propagation with macroscopic beachmarks, and Stage III eccentric fast overload rupture; (b) Microscopic cyclic striations advancing perpendicular to the local crack tip.

9 Conclusions

1. Rotating bending fatigue tests under fully reversed cyclic conditions ($R = -1$, $\sigma_m = 0$ MPa) were successfully conducted on Mild Steel and Aluminium alloy 6082-T6.
2. For Mild Steel, applied bending moments of 220 kg · cm ($d = 7.87$ mm) and 160 kg · cm ($d = 7.91$ mm) corresponded to alternating stress amplitudes of $\sigma_a = 450.84$ MPa and 322.93 MPa, resulting in fatigue lives of 16,400 and 46,722 cycles, respectively.

3. For Aluminium 6082, applied bending moments of 200 kg · cm ($d = 7.94$ mm) and 100 kg · cm ($d = 7.92$ mm) corresponded to stress amplitudes of $\sigma_a = 399.11$ MPa and 201.07 MPa, yielding fatigue lives of 3,991 and 154,585 cycles, respectively.

4. Our experimental dataset successfully spans both regimes: specimens MS-1, MS-2, and AL-1 fractured in **Low Cycle Fatigue (LCF)** ($N_f < 1 \times 10^5$ cycles), while specimen AL-2 at 201.07 MPa survived into **High Cycle Fatigue (HCF)** ($N_f = 154{,}585 > 1 \times 10^5$ cycles).

5. Mild steel demonstrated substantially superior fatigue endurance compared to aluminium 6082, surviving over 4× longer at higher stress levels (450.84 MPa vs 399.11 MPa).

6. Basquin power-law modeling on Aluminium 6082 extracted an exponent of $b_{\text{Al}} = -0.1875$ ($A_{\text{Al}} = 1{,}889.1$ MPa), predicting an arbitrary 10^7-cycle fatigue strength of $\sigma_{10^7} \approx 92.01$ MPa (29.7% σ_{UTS}), aligning with empirical structural alloy standards.

7. The fundamental difference is attributed to crystal structure: BCC mild steel benefits from interstitial Cottrell solute atmospheres that suppress cyclic dislocation motion, whereas FCC aluminium undergoes unpinned cross-slip on 12 slip systems, generating persistent slip bands without an endurance limit.

10 Appendix

10.1 Sample Stress and Moment Calculations

For Mild Steel Specimen MS-1 ($M = 220$ kg · cm, $d = 7.87$ mm):

$$M = 220 \text{ kg} \cdot \text{cm} \times 9.80665 \times 10^{-2} \text{ N} \cdot \text{m}/(\text{kg} \cdot \text{cm}) = \mathbf{21.5746} \text{ N} \cdot \text{m} \tag{20}$$

$$d = 7.87 \text{ mm} = 7.87 \times 10^{-3} \text{ m} \tag{21}$$

$$\sigma_a = \frac{32M}{\pi d^3} = \frac{32 \times 21.5746 \text{ N} \cdot \text{m}}{\pi \times (7.87 \times 10^{-3} \text{ m})^3} = \frac{690.388}{\pi \times 4.8744 \times 10^{-7}} \text{ Pa} = \mathbf{450.84} \text{ MPa} \tag{22}$$

$$\sigma_{\max} = +\sigma_a = +450.84 \text{ MPa}, \quad \sigma_{\min} = -\sigma_a = -450.84 \text{ MPa} \tag{23}$$

$$\sigma_m = \frac{\sigma_{\max} + \sigma_{\min}}{2} = \mathbf{0.0} \text{ MPa} \tag{24}$$

$$R = \frac{\sigma_{\min}}{\sigma_{\max}} = \frac{-450.84}{+450.84} = \mathbf{-1.0} \tag{25}$$

$$\Delta\sigma = \sigma_{\max} - \sigma_{\min} = 450.84 - (-450.84) = \mathbf{901.67} \text{ MPa} \tag{26}$$

$$\text{Regime:} \quad N_f = 16{,}400 < 1 \times 10^5 \text{ cycles} \implies \mathbf{Low\ Cycle\ Fatigue\ (LCF)} \tag{27}$$

For Aluminium Specimen AL-1 ($M = 200$ kg · cm, $d = 7.94$ mm, $N_f = 3{,}991$ cycles):

$$M = 200 \text{ kg} \cdot \text{cm} \times 9.80665 \times 10^{-2} \text{ N} \cdot \text{m}/(\text{kg} \cdot \text{cm}) = \mathbf{19.6133} \text{ N} \cdot \text{m} \tag{28}$$

$$d = 7.94 \text{ mm} = 7.94 \times 10^{-3} \text{ m} \tag{29}$$

$$\sigma_a = \frac{32M}{\pi d^3} = \frac{32 \times 19.6133 \text{ N} \cdot \text{m}}{\pi \times (7.94 \times 10^{-3} \text{ m})^3} = \frac{627.626}{\pi \times 5.0057 \times 10^{-7}} \text{ Pa} = \mathbf{399.11} \text{ MPa} \tag{30}$$

$$\Delta\sigma = 2\sigma_a = 2 \times 399.11 = \mathbf{798.21} \text{ MPa} \tag{31}$$

$$\text{Regime:} \quad N_f = 3{,}991 < 1 \times 10^5 \text{ cycles} \implies \mathbf{Low\ Cycle\ Fatigue\ (LCF)} \tag{32}$$

For Aluminium Specimen AL-2 ($M = 100$ kg $\cdot$ cm, $d = 7.92$ mm, $N_f = 154{,}585$ cycles):

$$M = 100 \text{ kg} \cdot \text{cm} \times 9.80665 \times 10^{-2} \text{ N} \cdot \text{m}/(\text{kg} \cdot \text{cm}) = \mathbf{9.8066} \text{ N} \cdot \text{m} \tag{33}$$

$$d = 7.92 \text{ mm} = 7.92 \times 10^{-3} \text{ m} \tag{34}$$

$$\sigma_a = \frac{32M}{\pi d^3} = \frac{32 \times 9.8066 \text{ N} \cdot \text{m}}{\pi \times (7.92 \times 10^{-3} \text{ m})^3} = \frac{313.811}{\pi \times 4.9679 \times 10^{-7}} \text{ Pa} = \mathbf{201.07} \text{ MPa} \tag{35}$$

$$\Delta\sigma = 2\sigma_a = 2 \times 201.07 = \mathbf{402.14} \text{ MPa} \tag{36}$$

$$\text{Regime:} \quad N_f = 154{,}585 > 1 \times 10^5 \text{ cycles} \implies \mathbf{High\ Cycle\ Fatigue\ (HCF)} \tag{37}$$

10.2 Rig Operation and Precautionary Notes

- Special spanners must be applied in diametrical pairs when tightening the collet chucks to ensure concentricity and prevent introducing pre-bending residual strains.
- The limit switch positioning directly beneath the loading hanger pan is verified before launching rotation to guarantee instantaneous motor cut-off and counter freezing upon specimen separation.
- Fracture surfaces should be stored in a dry desiccator to preserve the macroscopic beach-mark markings from atmospheric oxidation.